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விவசாயத் திணைக்களம்
Department of Agriculture



(<https://doa.gov.lk/>)

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யியல் பயிர் ஆராய்ச்சி அபிவிருத்தி நிலையம் - கன்னொறுவை
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(<https://www.facebook.com/pages/category/Government-Organization/Horticultural-Crops-Research-and-Development-Institute-274065813000866/>)

[<< Crop Page \(/hordi-crops/\)](/hordi-crops/)



Cucumber

Cucumis sativus

Cucumber is a popular vegetable as a cooking type or salad type

Released Varieties

- HORDI Green
- HORDI White

- Kalpitiya white **cu**
- Champion **cu**
- Gannoruwa white **m**
- LY 58 **be**

r/)
 (/Hordi-Variety-Cucumber/)
 Variety Details >>

Climatic requirements/ Areas suitable for cultivation

Suitable to cultivate in wet zone in year round and dry zone in Maha season. Can be cultivated up to above 1000 m sea level. Optimum temperature is 30 ° C.

Soil

Can be cultivated I with different soil types having better drainage and organic matter. Suitable pH is 5.5-7.5.

Seed requirement

1 kg/ha

Nursery Management

No nursery needed direct planting is practiced

Land preparation

After cleaning to remove weeds, hills can be prepared as 30m x30m x 30m in size.

Planting

Direct seeding

Spacing

1m x 1m as a spacing for mono culture. Sometime this crop can be cultivated as mix crop.

Fertilizer

Time of application	Urea kg/ha	TSP kg/ha	MOP kg/ha
Basal	75	200	60
After 4 weeks	75	–	60
After 5 weeks	75	–	60

Water supply

Apply sufficient amount of water, without lodging

Weed Control

Need to remove weeds at initial stage, better to use mulch to control weeds.

Harvesting

Yield

15-20 t/ha OP varieties

25-30 t/ha Hybrid varieties

Pest Management

This description will be available soon...

Till then, please access the Sinhala language page on insect control through below link.

View In Sinhala
(/Sinhala-Hordi-Crop-Cucumber)

Disease Management

- Fungal diseases

▲ 1. Powdery mildew

Causal organism: *Podosphaera xanthii* **Symptoms:**

- First appears as pale yellow spots on older leaves
- These spots enlarge as the white, fluffy mycelium grows over leaves surfaces
- Severely attacked leaves become brown and shrivelled and defoliation may occur

**Management:**

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)
- Destruction of infected plant debris after harvest

▲ [2. Downy mildew](#)

Causal organism: *Pseudoperonospora cubensis*

Symptoms:

- Angular yellow spots that are limited by the leaf veins on the upper surface of the leaves.
- Leaves become chlorotic and finally necrotic and host plant cells die

Management:

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)
- Distraction of infected plant debris after harvest

▲ 3. Collar rot/Root rot

Causal organisms: *Sclerotium* spp., *Fusarium* spp.

Symptoms:

- Initially the plants show temporary wilting symptoms
- The leaves yellowing, loose turgidity and show drooping
- Eventually, the plant dies.
- Vascular bundles in the collar region become yellow or brown.

Management:

- Destroy infected plants with soil
- Improved drainage
- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)

▲ 4. Gummy Stem Blight

Causal organisms: *Didymella bryoniae*

Symptoms:

- Lesions on leaves and fruit usually begin as spreading water-soaked areas
- The former these may have a chlorotic halo, become light brown and irregular in outline
- Dark cracked sunken lesions on fruit
- Plant collapse where sunken, girdling cankers
- Spots on stems often elongate into streaks and gummy exudates may occur from cracks
- Dark brown to black fruiting bodies (pycnidia) can be seen on fruit, stem, or leaf

Management:

- Crop debris should be ploughed deeply immediately after harvest to reduce fungus survival

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)

- **Bacterial diseases**

- **▲ 5. Bacterial wilt**

Causal organism: *Ralstonia solanacearum* **Symptoms:**

- Sudden drooping of leaves
- Whole plant finally wilts
- Leaves become brown, dry
- Vascular discoloration (Brown)
- Permanent wilting

Spreading of pathogen by;

- Infected soil, water, plant debris, Nematodes and insects ,pruning, Weeds as host plants



Disease Diagnosis: slimy viscous Bacterial ooze comes out from cut end of affected parts when immersing in clear water. **Management:**

- Destroy infected plants with soil
 - Crop rotation with non- susceptible crops (okra , maize)
 - Mixed cropping with cruciferae crops
 - Avoid movement of equipment from infected fields to non- infected fields
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- **Viral diseases**

- ▲ [6. Leaf Curl Virus \(LCV\)](#)

Transmitted by white fly (*Bemisia tabaci*) **Symptoms:**

- upward curling, shortening, and distortion of leaves
- Plant stunted and fruits deformed.



- ▲ [7. Papaya Ring Spot Virus \(PRSV\)](#)

Transmitted by aphids **Symptoms:**

- Mosaic and mottling of leaves

- leaf distortion
- Malformation of fruits and reduction in fruit size
- In severe infections, oily patches on the fruit surface



▲ [8. Cucumber Mosaic Virus \(CMV\)](#)

Transmission: Seed borne, through sap

Symptoms:

- Green mottle, leaf deformation
- Green spot on fruits, turn bright yellow and decaying fruit
- fruit malformation.

▲ [9. Zucchini Yellow Mosaic Virus \(ZYMV\)e](#)

Symptoms:

- Stunting, yellowing, necrosis (occasionally), mosaic, leaf deformations (blisters, shoe stringing)
- Fruit discolorations and deformations

• **Pytoplasma Disease**

▲ [Phytoplasma disease of cucurbits](#)

Disease symptoms

- Smaller, thickened and yellowish green leaves
- Shortened internodes and stunting of plants
- Witch's broom symptom

- Greening of flowers

Disease Transmission

- Through leaf hoppers
- Through infected seeds

Disease control

- Remove infected plants
- Control vectors (leaf hoppers) using a recommended insecticide

